

Estimation of High-Dimensional Copula-Based CUB Models using Composite Likelihood

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Abstract. Multivariate extensions of CUB (Combination of discrete Uniform and shifted Binomial) models via copulas face severe computational challenges due to the high number of terms required to evaluate the joint probability mass function. In this study, we propose a pairwise likelihood approach to overcome these limitations by focusing on bivariate marginal distributions. The methodology involves a two-step estimation procedure: individual bivariate models are fitted for each pair, and the resulting estimates are then summarized to obtain the parameters of the full joint model. Simulation results using the Gumbel copula demonstrate that the method is computationally efficient and provides accurate parameter estimates.

Keywords: CUB models, Copulas, Pairwise likelihood, Ordinal data

1 Introduction

In recent decades, research across various fields, including psychology, marketing, sociology, and behavioral sciences, has focused extensively on measuring and analyzing latent constructs such as perceptions, opinions, attitudes, and tastes. Information regarding these hidden traits is commonly gathered by administering questionnaires where respondents express their evaluations by selecting a score from a predefined set of ordered alternatives. The resulting data are typically classified as ordinal categorical variables, which pose significant statistical challenges. Their discrete and bounded nature requires specialized methods and models that can accommodate their unique properties, moving beyond the common but often inappropriate practice of treating them as numerical or continuous variables.

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Among the models proposed for rating data, the CUB (Combination of a discrete Uniform and a shifted Binomial random variable) class is particularly notable [13]. Originally proposed by Piccolo and D’elia [6, 12], this framework posits that the final rating is the outcome of a joint latent decision process involving two distinct components: feeling and uncertainty. Statistically, these are modelled through a mixture distribution, where a shifted Binomial random variable models the individual’s subjective sensation or agreement, the feeling; while a discrete Uniform distribution accounts for the inherent indecision and potential randomness in the choice process, the uncertainty.

Formally, the CUB model is defined as follows:

$$P(R = r \mid \pi, \xi) = \pi \binom{m-1}{r-1} \xi^{m-r} (1-\xi)^{r-1} + (1-\pi) \frac{1}{m}, \quad r = 1, \dots, m \quad (1)$$

where m is the number of levels, $1 - \xi$ is the feeling parameter, and $1 - \pi$ is the uncertainty parameter.

While the univariate CUB model has been widely extended to include covariates and different response styles, its extension to a multivariate setting remains a complex frontier. In practice, it is common to collect multiple ratings from the same individual across several items, leading to multivariate ordinal data where the dependence between responses must be appropriately modeled. Since this is a complex task, some authors proposed multivariate CUB models assuming independence among the variables, to easily fit the model in a model-based clustering context [15]. However, the idea of coupling CUB models with copulas is not new. In the bivariate context, Andreis and Ferrari [1] explored the use of the Archimedean family to capture the dependence between two CUB random variables, while Corduas [3, 4] proposed an approach based on the Plackett distribution to model the association between ordinal responses, also allowing for the inclusion of subject-specific covariates. Colombi and Giordano [2] proposed a bivariate construction using the Sarmanov approach. For a multivariate extension, Corduas [5] utilized a Plackett distribution as a copula together with a discrete vine pair-copula construction to achieve computational efficiency. In contrast, the model suggested by Ip and Wu [9] is very different, extending the mixture of a binomial and a uniform distribution to two dimensions through a mixture of bivariate discrete uniforms and bivariate binomials. However, as the number of dimensions d increases, the full likelihood estimation for copula-based CUB models becomes computationally intensive due to the 2^d terms required to evaluate the joint probability mass functions.

In this paper, we address these severe computational limitations by proposing a pairwise likelihood approach. This methodology simplifies the estimation process by focusing exclusively on bivariate marginal distributions, which provides a robust and efficient alternative for modeling high-dimensional dependency structures in rating data.

2 Modelling complex interdependencies with Copulas

The complexity of defining a multivariate CUB model through copulas stems primarily from the discrete and ordinal nature of the data. While the use of a d -dimensional copula for continuous variables is computationally straightforward, in the case of discrete variables, the joint probability mass function must be derived through finite differences of the joint distribution function. Following Panagiotelis et al. [11], the probability of observing a vector of responses $\mathbf{r} = (r_1, \dots, r_d)'$ with $r_1 \in \{1, \dots, m_1\}, r_d \in \{1, \dots, m_d\}$, is given by:

$$P(\mathbf{r}) = \sum_{\mathbf{v} \in \mathcal{V}} \text{sgn}(\mathbf{v}) C(G_1(v_1), \dots, G_d(v_d)), \quad (2)$$

where each $G_j(\cdot)$ represents the marginal cumulative distribution function of a CUB model with parameters (π_j, ξ_j) and $C(\cdot, \dots, \cdot)$ is any d -dimensional copula function. The vector $\mathbf{v} = (v_1, \dots, v_d)'$ represents the vertices of the unit hypercube defined by $v_j \in \{r_j, r_j - 1\}$, and the sign function is defined as:

$$\text{sgn}(\mathbf{v}) = \begin{cases} 1, & \text{if } v_j = r_j - 1 \text{ for an even number of indices } j \\ -1, & \text{if } v_j = r_j - 1 \text{ for an odd number of indices } j \end{cases}. \quad (3)$$

Following Panagiotelis et al. [11], the joint probability of observing a vector of responses $\mathbf{r}$ is obtained by evaluating the copula at the vertices of a d -dimensional unit hypercube. This involves the summation of 2^d terms, as shown in Equation 2. Consequently, the Full Likelihood (FL) is defined as the product of these joint probabilities over the sample of n independent observations and its logarithm is given by:

$$\ell_{FL}(\boldsymbol{\theta}) = \sum_{i=1}^n \ln P(r_{i1}, \dots, r_{id}; \pi_1, \dots, \pi_d, \xi_1, \dots, \xi_d, \boldsymbol{\alpha}) \quad (4)$$

where $\boldsymbol{\theta}$ denotes the vector of all CUB (π, ξ) and copula parameters $(\boldsymbol{\alpha})$.

For a model in dimension d , the evaluation of ℓ_{FL} becomes extremely burdensome. Consider, for instance, a psychometric questionnaire with $d = 10$ items: the estimation would require calculating $2^{10} = 1024$ terms for every single observation at each iteration of the maximization process. In high-dimensional settings, this makes the maximization of the usual likelihood often prohibitive in terms of time and computational resources.

2.1 A pairwise likelihood approach for multivariate CUB models with copulas

The fundamental idea that allow us to overcome these difficulties is that, rather than defining and estimating the full d -dimensional model, it may be sufficient to consider only the bivariate marginal distributions. To do that, we propose to

adopt a pairwise likelihood (PL) approach [10, 14], which for a d -dimensional model is defined as follows:

$$\ell_{PL}(\boldsymbol{\theta}_{jk}) = \sum_{i=1}^n \sum_{j < k} \ln P(r_{ij}, r_{ik}; \pi_j, \pi_k, \xi_j, \xi_k, \alpha_{jk}) \quad (5)$$

where each term $P(r_{ij}, r_{ik}; \pi_j, \pi_k, \xi_j, \xi_k, \alpha_{jk})$ is the bivariate distribution defined by CUB marginals for the j -th and k -th item, and a chosen copula with dependence parameter α_{jk} .

This formulation clearly shows that each component of the sum depends only on the parameters of two items at a time, significantly reducing the computational complexity compared to the full likelihood.

This approach offers a two-fold advantage: it drastically reduces computational complexity, limiting the calculation to only $2^2 = 4$ terms per pair, and it allows for the use of a wide range of copula families to model different types of interdependencies between pairs of variables, an operation that would be far more complex in a purely multivariate setting. However, the simultaneous optimization of $\ell_{PL}(\theta)$ can still be challenging as the number of parameters grows. Following Fieuwis and Verbeke [7], we adopt a two-step estimation procedure. In the first step, we independently estimate each of the $d(d-1)/2$ bivariate models:

$$\hat{\theta}_{jk} = \arg \max_{\theta_{jk}} \sum_{i=1}^n \ln P(r_{ij}, r_{ik}; \pi_j, \pi_k, \xi_j, \xi_k, \alpha_{jk}). \quad (6)$$

It is important to note that the parameter vector θ_{jk} contains both the marginal parameters of the CUB models $(\pi_j, \xi_j, \pi_k, \xi_k)$ and the copula dependence parameter α_{jk} . Consequently, each marginal parameter appears in $d-1$ different bivariate models. In the above setting, α_{jk} denotes the association parameter specific to the pair of responses (j, k) . By default, we allow for a saturated association structure where each pair is characterized by a unique α_{jk} . However, for the sake of parsimony, equality constraints can be imposed (e.g., $\alpha_{jk} = \alpha$ for all $j \neq k$) to reduce the number of parameters to be estimated, particularly in high-dimensional settings. In the simulation study, the equality constraint $\alpha_{jk} = \alpha$ for all $j \neq k$ is adopted for parsimony. At the second stage, to obtain a unique estimate for each parameter of the full joint model, we summarize the obtained estimates. The most effective solution consists of calculating the average (simple or weighted by the Hessian matrix) of the individual estimates for each parameter across all its occurrences in the bivariate models. Note that standard errors for the full pairwise likelihood method can be estimated via the Godambe information matrix [14], while for our approach with the method described in [8].

3 Simulation study

To evaluate the performance of the proposed pairwise likelihood approach and compare it with the standard full likelihood estimation, we carried out two dis-

tinct simulation experiments. The first experiment aims at evaluating the accuracy of estimates and the efficiency of the PL relative to the FL approach, while the second assesses computational scalability as the number of items increases.

In all scenarios, data were generated from a multivariate CUB distribution coupled with three Archimedean copula families: Clayton, Frank, and Gumbel. These families were selected to represent a broad range of dependence structures: the Clayton copula exhibits lower-tail dependence, the Gumbel copula upper-tail dependence and the Frank copula symmetric dependence with no tail concentration.

3.1 Parameter estimates and efficiency

We considered a set of 5 items, since for $d = 5$ the full likelihood approach begins to reach its practical limit for intensive simulations due to the $2^5 = 32$ terms required for each joint probability evaluation. To focus the comparison on the effect of copula family and dependence strength, the marginal parameters were set equal across all items: $\pi_j = 0.6$ and $\xi_j = 0.7$ for $j = 1, \dots, 5$. Dependence strength was calibrated in terms of Kendall's τ , set to 0.3 (weak) and 0.6 (strong), and converted into the copula-specific parameter for each family. We tested sample sizes $n \in \{100, 500, 1000\}$ with $B = 100$ replicates for each scenario. In the second stage of the estimation procedure, the marginal parameter estimates are synthesised via averaging across all bivariate models in which each parameter appears. The performance was evaluated through the estimates' mean, standard deviation (SD), and mean squared error (MSE), which is computed for each parameter θ as the average of the squared differences between the estimates $\hat{\theta}_b$ obtained in each of the B replications and the true parameter value, namely $MSE(\hat{\theta}) = \frac{1}{B} \sum_{b=1}^B (\hat{\theta}_b - \theta)^2$. By including this metric, we account for both the precision and the accuracy of the estimates. Both PL and FL are consistent across all copula families considered, with MSE values decreasing as n grows. A slightly higher variability in the estimation of π parameters compared to ξ is observed, a known characteristic of the CUB family arising from the mixture nature of the model. To assess the loss of efficiency when moving from FL to PL, we calculated the relative efficiency (RE), defined as $RE = MSE_{FL}/MSE_{PL}$. The results, reported in Figure 1, show that RE values remain consistently close to 1 across all copula families and dependence levels, confirming that the pairwise approach incurs negligible loss of estimation accuracy regardless of the dependence structure.

Standard errors for the PL estimator were obtained via the Godambe information matrix following [8], while for the FL estimator they were derived from the observed Hessian. To assess the reliability of the uncertainty quantification, we compared, for each scenario, the average estimated Standard Errors with the empirical Standard Deviations across replications. Table 1 reports this comparison at $n = 500$ for π_1 , ξ_1 , and α across all copula families and dependence levels. The two quantities are in close agreement throughout, indicating that the Standard Errors estimator proposed in [8] provides well-calibrated standard errors for the PL approach.

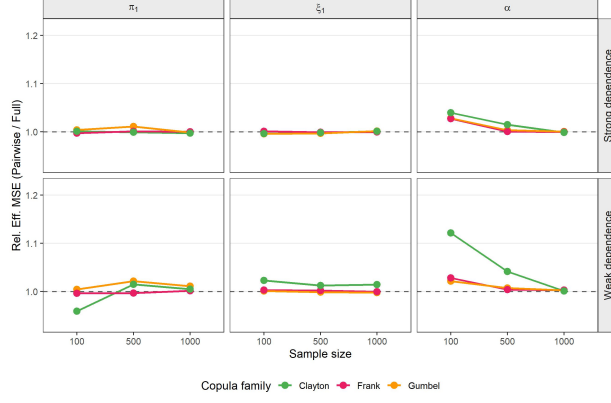


Fig. 1. Relative efficiency for the $d = 5$ multivariate CUB model across copula families (Clayton, Frank, Gumbel), dependence levels (weak: $\tau = 0.3$, strong: $\tau = 0.6$), and sample sizes $n \in \{100, 500, 1000\}$. Values close to 1 indicate negligible loss of efficiency of PL relative to FL.

Table 1. Empirical standard deviations (SD) and average estimated standard errors (SE) for PL and FL at $n = 500$, across copula families and dependence levels, for parameters π_1 , ξ_1 , and α .

		Gumbel		Frank		Clayton	
Param		Weak	Strong	Weak	Strong	Weak	Strong
<i>Pairwise Likelihood (PL)</i>							
π_1	SD	0.0541	0.0559	0.0618	0.0530	0.0547	0.0602
	SE	0.0542	0.0534	0.0551	0.0535	0.0552	0.0538
ξ_1	SD	0.0211	0.0188	0.0183	0.0174	0.0204	0.0211
	SE	0.0189	0.0185	0.0188	0.0172	0.0187	0.0180
α	SD	0.0401	0.0974	0.1730	0.3192	0.0746	0.2152
	SE	0.0401	0.1000	0.1793	0.3232	0.0683	0.1907
<i>Full Likelihood (FL)</i>							
π_1	SD	0.0511	0.0532	0.0615	0.0517	0.0541	0.0598
	SE	0.0521	0.0514	0.0547	0.0518	0.0550	0.0530
ξ_1	SD	0.0206	0.0164	0.0174	0.0152	0.0196	0.0206
	SE	0.0184	0.0175	0.0181	0.0159	0.0183	0.0170
α	SD	0.0349	0.0940	0.1670	0.3151	0.0691	0.2078
	SE	0.0372	0.0960	0.1711	0.3122	0.0655	0.1794

3.2 Computational scalability

The second experiment aimed at quantifying the computational gain offered by the pairwise likelihood approach.

We measured the execution time as a function of the model's dimensionality, ranging from $d = 2$ to $d = 15$, using a single replication per scenario with $m = 5$ categories for each ordinal variable and dependence calibrated at $\tau = 0.6$. A timeout of 10 minutes was imposed for each replication: when FL estimation exceeded this threshold the curve was interrupted, which explains why the FL lines terminate before $d = 15$ for all families.

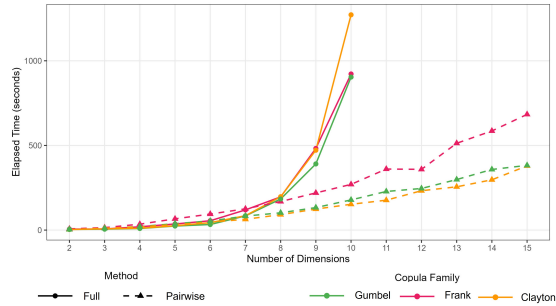


Fig. 2. Execution time (seconds) for FL vs PL as a function of the dimension d . An upper limit of 10 minutes was used.

The results reported in Figure 2 consistently highlight the curse of dimensionality affecting the joint full likelihood across all Archimedean copula families considered. The FL execution time grows steeply on the log scale, reaching the timeout threshold at $d = 10$ for all the families, whereas the PL curves continue smoothly up to $d = 15$ in all cases. The two methods perform comparably up to $d = 8$, after which the FL execution time increases sharply, reaching over 900 seconds at $d = 10$ for all families while the PL method remains below 250 seconds. The observed computational times are consistent with expectations, as they reflect the exponential growth in the number of terms for the first model and the quadratic growth for the second.

4 Discussion

In this study, we proposed a multivariate CUB model framework that accounts for interdependencies through copula functions. To address the computational complexity inherent in high-dimensional settings, we introduced an innovative approach based on pairwise likelihood estimation.

Preliminary simulation studies demonstrate that this method offers significant computational advantages as the number of dimensions increases, while maintaining high estimation accuracy. Specifically, it reveals only a negligible loss of efficiency in parameter estimation when compared to the full likelihood approach. The flexibility of the proposed method is a key strength, as it accommodates different numbers of categories for each dimension and supports various copula families to capture diverse dependency structures.

Future research will involve more extensive simulation studies to further validate the model's robustness. We also note that, while the proposed framework supports a saturated association structure with pair-specific parameters, the simulation study adopts a common dependence parameter across pairs for parsimony; this parameter is estimated separately for each bivariate model and then synthesised into a single final estimate. The proposed framework can be extended to accommodate richer dependence structures by adopting multivariate copulas

with more flexible parameterisations, such as the Gaussian copula. Additionally, we plan to apply this procedure to data from the European Social Survey, where multiple thematic modules collect batteries of Likert-scale items across a large number of respondents, representing a natural high-dimensional setting for the proposed methodology.

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